Module 12

Latent Variabl Model

Kentaro Nakamura

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Agenda

- k-means algorithm

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- Gaussian Mixture Models (GMM)

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- EM Algorithm in general
  - How to derive E / M-step
  - Why EM algorithm works
  - Uncertainty quantification under EM

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- Gaussian Mixture Models (GMM)
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  - Why EM algorithm works
  - Uncertainty quantification under EM
- Useful resource: Bishop (2006) Chapter 9
- For future reference (not for final exam, but in appendix)
  - Variational Inference
  - MCMC Methods

k-means algorithm (Overview)

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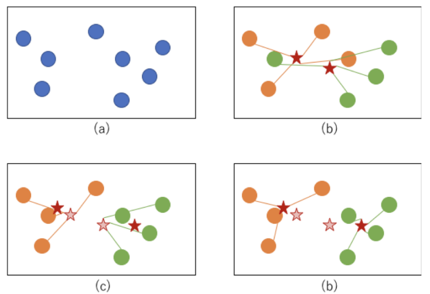

k-means algorithm: optimization (1)

- Formally, the objective function is written as

$$\{\boldsymbol{Z}, \boldsymbol{\mu}\} := \arg\min_{\boldsymbol{Z}, \boldsymbol{\mu}} \sum_{i=1}^{n} \sum_{k=1}^{K} \mathbf{1}\{Z_i = k\} ||\boldsymbol{X}_i - \boldsymbol{\mu}_k||_2^2$$

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- Now, let's fix the assignment $\boldsymbol{Z}$. Then, the derivative w.r.t. μ_k is

$$\frac{\partial}{\partial \mu_k} \sum_{i:Z_i=k} ||\boldsymbol{X}_i - \boldsymbol{\mu}_k||_2^2 = -2 \cdot \sum_{i:Z_i=k} (\boldsymbol{X}_i - \mu_k)$$

  which means that setting $\mu_k = \frac{1}{n_k} \sum_{i:Z_i=k} \boldsymbol{X}_i$ (equivalent to STEP 2 in Lloyd's algorithm) achieves the first order condition

k-means algorithm: optimization (2)

- Now, think about optimizing cluster assignment given the centroid

$$\{\boldsymbol{Z}^{(t+1)}\} := \arg\min_{\boldsymbol{Z}} \sum_{i=1}^{n} \sum_{k=1}^{K} \mathbf{1}\{Z_i = k\} ||\boldsymbol{X}_i - \boldsymbol{\mu}_k^{(t)}||_2^2$$

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- Note that the objective function is separable across i:

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- This is exactly STEP 3 in Lloyd's algorithm (assign to nearest centroid)
  - Note that this might yield local optimal depending on initialization, so typically repeat the procedure with multiple initialization and pick up the one that minimizes the objective function the most.

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  where $\boldsymbol{\pi} = [\pi_1, \cdots, \pi_K]^\top$, $\sum_{k=1}^{K} \pi_k = 1$ and $0 \leq \pi_k \leq 1$.

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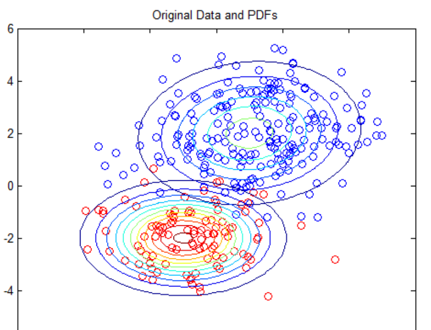

Likelihood of Latent Variable Model (1)

- Let's start with the likelihood of GMM as if we observe latent variable.

$$\mathcal{L}^{(\text{full})}(\boldsymbol{X}, \boldsymbol{Z}; \boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \prod_{i=1}^{n} \prod_{k=1}^{K} \left(\pi_k \mathcal{N}(X_i; \mu_k, \Sigma_k)\right)^{\mathbf{1}\{Z_i = k\}}$$

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- Thus, if we observe Z_i, then we get MLE

$$\hat{\pi}_k = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{Z_i = k\}, \qquad \hat{\mu}_k = \frac{\sum_{i=1}^{n} \mathbf{1}\{Z_i = k\} X_i}{\sum_{i=1}^{n} \mathbf{1}\{Z_i = k\}}$$

$$\hat{\Sigma}_k = \frac{1}{\sum_{i=1}^{n} \mathbf{1}\{Z_i = k\}} \sum_{i=1}^{n} \mathbf{1}\{Z_i = k\}(X_i - \hat{\mu}_k)(X_i - \hat{\mu}_k)^\top$$

  - So, if we observe Z_i, then the problem is easy

Likelihood of Latent Variable Model (2)

- However, Z_i is latent. We cannot observe. What we observe is the observed likelihood:

$$
\begin{aligned}
\mathcal{L}^{(\mathrm{obs})}(\boldsymbol{X}; \boldsymbol{\mu}, \boldsymbol{\Sigma}, \boldsymbol{\pi}) &= \sum_{\boldsymbol{z}} \mathcal{L}^{(\mathrm{full})}(\boldsymbol{X}, \boldsymbol{z}; \boldsymbol{\mu}, \boldsymbol{\Sigma}, \boldsymbol{\pi}) \\
&= \sum_{z_1, \cdots, z_n} \prod_{i=1}^{n} \prod_{k=1}^{K} \left(\pi_k \mathcal{N}(X_i; \mu_k, \Sigma_k)\right)^{\mathbf{1}\{Z_i = k\}} \\
&= \sum_{z_1, \cdots, z_n} \prod_{i=1}^{n} \left(\pi_{Z_i} \mathcal{N}(X_i; \mu_{Z_i}, \Sigma_{Z_i})\right) \\
&= \prod_{i=1}^{n} \sum_{z_i} \left(\pi_{Z_i} \mathcal{N}(X_i; \mu_{Z_i}, \Sigma_{Z_i})\right) \\
&= \prod_{i=1}^{n} \sum_{k} \pi_k \mathcal{N}(X_i; \mu_k, \Sigma_k)
\end{aligned}
$$

where the last equality is by re-indexing.

  - Recall that $\Pr(X = x) = \sum_y \Pr(X_i = x, Y_i = y)$ (this is why you need to integrate out the latent)

Problem in Latent Variable Model

- Therefore, the observed data log likelihood is given by

$$\log p(\boldsymbol{X} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}, \boldsymbol{\pi}) = \sum_{i=1}^{n} \log \left\{ \sum_{k} \pi_k \mathcal{N}(X_i; \mu_k, \Sigma_k) \right\}$$

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- This is hard to maximize because of the structure log (sum)
  - The log of sum over convex function is generally not convex
  - To see why, consider the example

    $$f(x) = \log\{\exp(-[x-2]^2) + \exp(-([x+2]^2)\}$$

    where you see each term is convex
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- Similarly, GMM suffers from non-convexity problem
  - There are K latent cluster, but each cluster label does not have meaning
  - **Label switching**: There are $K!$ equivalent solutions

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- Now, check the first order condition for each parameter. The derivative w.r.t. $\boldsymbol{\mu}_k$ is given by

$$0 = -\sum_{i=1}^{n} \underbrace{\frac{\pi_k \mathcal{N}(\boldsymbol{X}_i; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(\boldsymbol{X}_i; \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}}_{:=\gamma_{i,k}(=\Pr(Z_i=k|\boldsymbol{X}_i))} \boldsymbol{\Sigma}_k^{-1}(X_i - \boldsymbol{\mu}_k)$$

where $\gamma_{i,k}$ is called responsibility parameter.

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- Similarly, the derivative with respect to $\boldsymbol{\mu}_k$ yields

$$\boldsymbol{\mu}_k = \frac{\sum_{i=1}^{n} \gamma_{i,k} \boldsymbol{X}_i}{\sum_{i=1}^{n} \gamma_{i,k}}$$

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&= \frac{1}{2} \sum_{i=1}^{n} \gamma_{i,k} \left[\boldsymbol{\Sigma}_k - (\boldsymbol{X}_i - \boldsymbol{\mu}_k)(\boldsymbol{X}_i - \boldsymbol{\mu}_k)^\top \right].
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which yields

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- Since $\sum \pi_k = 1$, $\sum_k \{\sum_{i=1}^{n} \gamma_{i,k} + \lambda \pi_k\} = \sum_k \sum_{i=1}^{n} \gamma_{i,k} + \lambda = 0$.
  - I.e., $\lambda = -N$
  - This gives us $\pi_k = \sum_{i=1}^{n} \gamma_{i,k} / N$.

EM Algorithm for GMM (4)

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- **E-Step**: Calculate the responsibility parameter given $\boldsymbol{\mu}_k$, $\boldsymbol{\Sigma}_k$, and $\boldsymbol{\pi}$:

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- **M-Step**: Maximize the log-likelihood. I.e., obtain the MLE estimator given $\gamma_{i,k}$ by

$$\boldsymbol{\mu}_k = \frac{\sum_{i=1}^{n} \gamma_{i,k} \boldsymbol{X}_i}{\sum_{i=1}^{n} \gamma_{i,k}}, \quad \boldsymbol{\Sigma}_k = \frac{\sum_{i=1}^{n} \gamma_{i,k} (\boldsymbol{X}_i - \boldsymbol{\mu}_k)(\boldsymbol{X}_i - \boldsymbol{\mu}_k)^\top}{\sum_{i=1}^{n} \gamma_{i,k}}$$

$$\pi_k = \frac{\sum_{i=1}^{n} \gamma_{i,k}}{N}$$

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- **Setup**: Let $\boldsymbol{x}$ be the observed data, $\boldsymbol{z}$ be latent, and θ be the parameter of the model
  - Then, the full-data log likelihood is written as $\log p(\boldsymbol{x}, \boldsymbol{z}, \theta)$ and observed data log likelihood (known as evidence) is $\log p(\boldsymbol{x}; \theta)$

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  - EM avoids this by:
    - **E-step**: "fill in" missing data using current model (soft assignment)
    - **M-step**: maximize as if data were complete

Theory of EM Algorithm (1)

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$$
\begin{aligned}
\log p(x;\theta) &= \log\Big(\sum_z p(x,z;\theta)\Big) \\
&= \log\Big(\sum_z \frac{p(x,z;\theta)}{q(z)} q(z)\Big) \\
&\geq \sum_z \log \frac{p(x,z;\theta)}{q(z)} q(z) \quad (\because \text{ Jensen's inequality}) \\
&= \mathbb{E}_z[\log p(x,z;\theta)] - \mathbb{E}_z[\log q(z)] := \mathcal{L}(q,\theta)
\end{aligned}
$$

where the last term $\mathcal{L}(q,\theta)$ is called **Evidence Lower Bound** (ELBO)

Theory of EM Algorithm (2)

- Notice that ELBO is written as

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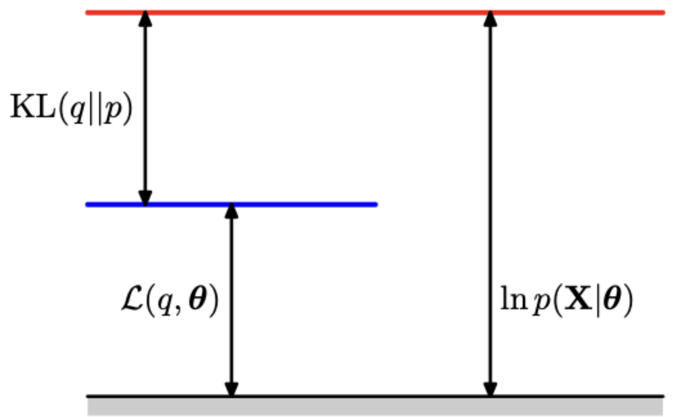
- Recall that KL-divergence is non-negative and takes 0 when $q(z) = p(z \mid x;\theta)$

Theory of EM Algorithm (3)



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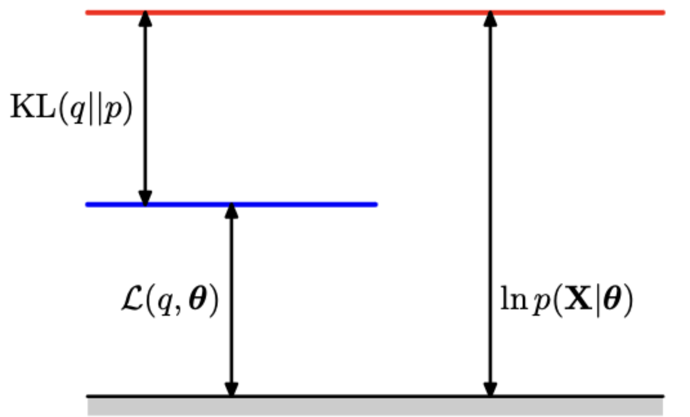

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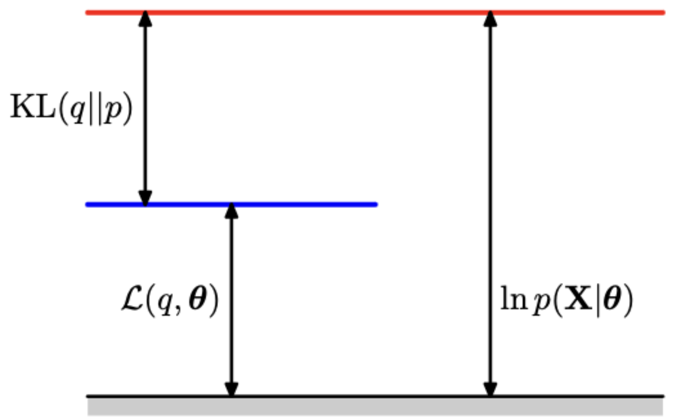

- Recall that our goal is to maximize the observed log-likelihood $\log p(x; \theta)$
  - This corresponds to maximize the lower bound $\mathcal{L}(q, \theta)$
- Notice that we have two unknowns: q and θ
  - We iteratively update them

Theory of EM Algorithm (4)

- **E-step**: Maximize ELBO $\mathcal{L}(q, \theta)$ with respect to q while fixing $\theta = \theta^{\text{old}}$.
    - Notice that once θ is fixed, we can maximize ELBO by choosing q makes KL divergence to be 0 (recall the previous derivation)

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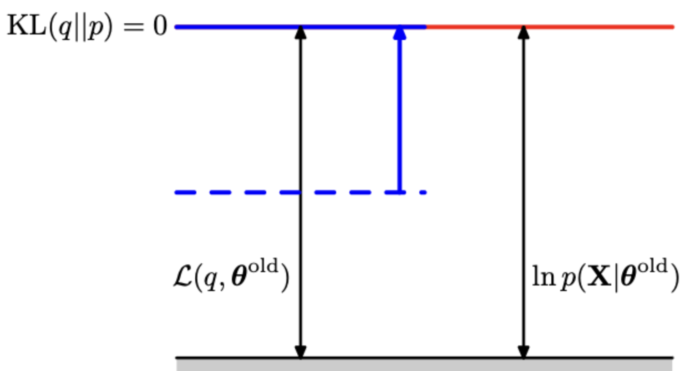

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- This is the justification of maximizing Q-function

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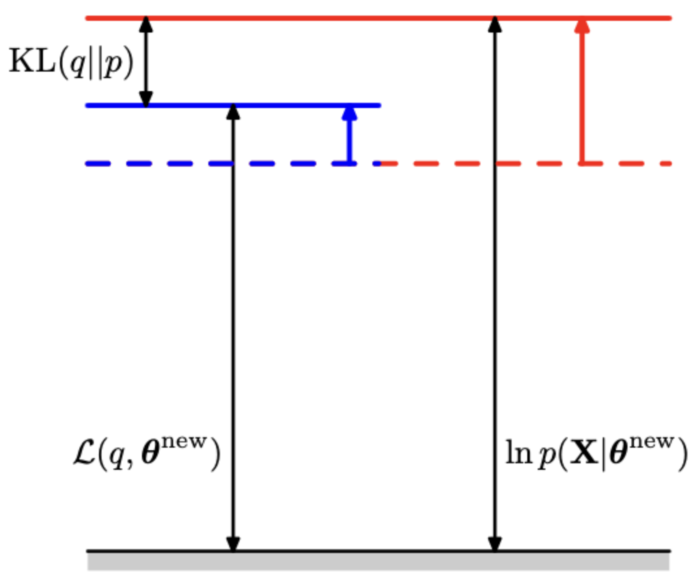

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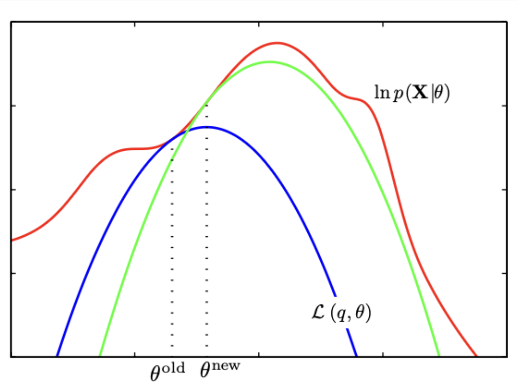

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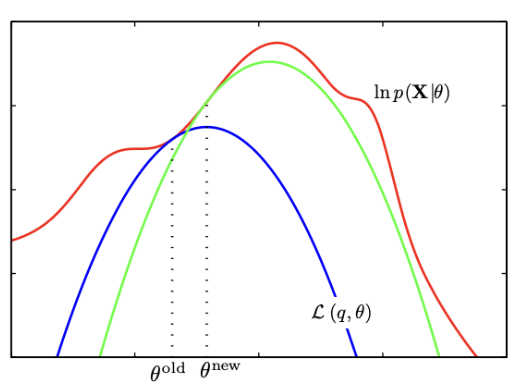

- E-step: Make ELBO $\mathcal{L}(q, \theta)$ tangential to log-likelihood (maximization of ELBO w.r.t. q under fixed θ)
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- Thus, the observed-data log-likelihood never decreases.

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  - Implication: We can easily obtain the complete data information (similar to E-step), so we only need to obtain missing data part
    - Importantly, this avoids differentiating the observed log-likelihood directly (which involves log-sum structure)

Derivation of Missing-Data Principle

- Now, by factorization,

$$f(\boldsymbol{X}, \boldsymbol{Z} \mid \theta) = f(\boldsymbol{Z} \mid \boldsymbol{X}, \theta) f(\boldsymbol{X} \mid \theta).$$

- So taking logs gives

$$\ell(\theta \mid \boldsymbol{X}, \boldsymbol{Z}) = \log f(\boldsymbol{X}, \boldsymbol{Z} \mid \theta) = \log f(\boldsymbol{Z} \mid \boldsymbol{X}, \theta) + \log f(\boldsymbol{X} \mid \theta).$$

- Now take the Hessian with respect to θ, we get

$$\ell''(\theta \mid \boldsymbol{X}) = \ell''(\theta \mid \boldsymbol{X}, \boldsymbol{Z}) - \nabla^2 \log f(\boldsymbol{Z} \mid \boldsymbol{X}, \theta).$$

- Finally, take conditional expectation given $\boldsymbol{X}$ (notice that the first term only depends on $\boldsymbol{X}$)

$$-\ell''(\theta \mid \boldsymbol{X}) = \mathbb{E}[-\ell''(\theta \mid \boldsymbol{X}, \boldsymbol{Z}) \mid \boldsymbol{X}] - \mathbb{E}\left[-\nabla^2 \log f(\boldsymbol{Z} \mid \boldsymbol{X}, \theta) \mid \boldsymbol{X}\right].$$

Extra: Approximation Methods

- **Problem of EM algorithm**: We need to evaluate the expectation of the complete-data log-likelihood with respect to the posterior distribution of latent variable
  - In many applications, however, this is not possible.
- In such situation, we need to use approximation methods
  - Stochastic: MCMC (Metropolis Hasting, Gibbs Sampling, Hamiltonian Monte Carlo)
    - Gives accurate estimates with uncertainty quantification, but not scalable for large and complex data
  - Deterministic: Variational Inference
    - Gives approximate estimates, but scalable and fast
- This is not required for final exam, but it is the main technique you need to use in future
  - Variational Inference: Ch.10 of Bishop (2006), Blei et al. (2017 JASA)
  - MCMC: Ch.11 of Bishop (2006)

Extra: Variational Inference (1)

- Now, we are in the bayesian framework. So, the parameter θ is a random variable
  - Thus, we denote the set of all parameters and latent variable as $\boldsymbol{Z}$
  - Our goal is to find the distribution that is the closest to the exact posterior $p(\boldsymbol{Z} \mid \boldsymbol{X})$, which is minimizing the KL-divergence $\mathrm{KL}(q(z) \mid p(z \mid x))$ over some candidate set of distributions $\mathcal{Q}$.

- Importantly, the minimization of KL divergence is equivalent to the maximization of ELBO:

$$
\begin{aligned}
q^*(z) &= \arg\min_{q(z)\in\mathcal{Q}} \mathrm{KL}(q(z) \,||\, \underbrace{p(z \mid x)}_{\text{Posterior}}) \\
&= \arg\min_{q(z)\in\mathcal{Q}} \left(\underbrace{\log p(x)}_{\text{constant}} - \mathrm{ELBO}(q) \right) \\
&= \arg\max_{q(z)\in\mathcal{Q}} \mathrm{ELBO}(q)
\end{aligned}
$$

Extra: Variational Inference (2)

- What kind of $\mathcal{Q}$ should we use?
  - **Mean Field Approximation**: Choose $\mathcal{Q}$ to restrict that latent variables are mutually independent; i.e.,

$$q(z) = \prod_{j=1}^{m} q_j(z_j)$$

  - It is crazy assumption, but often works well in practice

- **Pro**: Mean-field family is expressive because it can capture any marginal density of the latent variable
- **Con**: It cannot capture the correlation between latent variables

Extra: Variational Inference (3)

- Now, let's focus on j-th latent $q_j(z_j)$.

$$
\begin{aligned}
&\mathrm{ELBO}(q_j) \\
&= \mathbb{E}_{z_j \sim q_j}\Big[\underbrace{\mathbb{E}_{z_{-j}}\Big\{\log p(x, z_j, z_{-j})\Big\}}_{:= f(z_j)}\Big] - \mathbb{E}_{z_j \sim q_j}[\log q_j(z_j)] + \mathrm{const} \\
&= -\mathbb{E}_{z_j \sim q_j}\left[\log \frac{q_j(z_j)}{\exp(f(z_j))}\right] + \mathrm{const} \\
&= -\mathrm{KL}(q_j(z_j) \,||\, \exp(f(z_j))) + \mathrm{const}
\end{aligned}
$$

- Thus, when ELBO is maximized,

$$
q_j^*(z_j) \propto \exp\Big\{\mathbb{E}_{z_{-j}}\Big\{\log p(x, z_j, z_{-j})\Big\}\Big\}
$$

Extra: Variational Inference (4)

Coordinate ascent mean-field variational inference (CAVI)

- **Input**: A model $p(x, z)$, data $p(z)$
- **Output**: Variational density $q(z)$
- **Algorithm**

  1. for each $j \in 1, \cdots, m$, set

  $$q_j(z_j) \propto \exp\left\{\mathbb{E}_{z_{-j}}\left\{\log p(x, z_j, z_{-j})\right\}\right\}$$

     - Each coordinate density is conditional on all the other coordinates
     - Similar to Gibbs sampling

  2. Compute $\mathrm{ELBO}(q)$ and repeat until convergence

- **Caveat**: ELBO is a non-convex objective function, which is only guaranteed to converge to a local optimum

$\rightarrow$ Sensitive to the initialization

Extra: From VI to MCMC

- In both VI and MCMC, the goal is often to approximate a complicated distribution

- In Bayesian latent variable models, this target is usually the posterior
$$p^*(\boldsymbol{z}) = p(\boldsymbol{z} \mid \boldsymbol{X}).$$

- But for MCMC theory, we do not need to write the observed data $\boldsymbol{X}$ explicitly

- So, from now on, we simply write the target distribution as
$$p^*(\boldsymbol{z}).$$

- **VI**: approximate $p^*(\boldsymbol{z})$ by optimizing over $q(\boldsymbol{z})$

- **MCMC**: generate samples from $p^*(\boldsymbol{z})$ by constructing a Markov chain whose stationary distribution is $p^*(\boldsymbol{z})$

Extra: Theory of MCMC (1)

- A first-order Markov chain is defined as

$$p(\boldsymbol{z}^{m+1} \mid \boldsymbol{z}^1, \cdots, \boldsymbol{z}^m) = p(\boldsymbol{z}^{m+1} \mid \boldsymbol{z}^m)$$

- Thus, by the law of total probability,

$$p(\boldsymbol{z}^{m+1}) = \sum_{\boldsymbol{z}^m} \underbrace{p(\boldsymbol{z}^{m+1} \mid \boldsymbol{z}^m)}_{\text{Transition Prob}} p(\boldsymbol{z}^m)$$

- **Detailed Balancing**: With transition probabilities $T(\boldsymbol{z}, \boldsymbol{z}')$,

$$p^*(\boldsymbol{z}) T(\boldsymbol{z}, \boldsymbol{z}') = p^*(\boldsymbol{z}') T(\boldsymbol{z}', \boldsymbol{z})$$

- Detailed balancing is a *sufficient* condition to ensure that the required distribution $p(\boldsymbol{z})$ is stationary

Extra: Theory of MCMC (2)

- The detailed balancing implies stationarity because of the following:

$$
\begin{aligned}
p(z) &= \sum_{z'} p^{prev}(z') T(z', z) \\
&= \sum_{z'} p^{prev}(z) T(z, z') \quad \text{(detailed balancing)} \\
&= p^{prev}(z) \sum_{z'} T(z, z') \\
&= p^{prev}(z) \underbrace{\sum_{z'} p(z' \mid z)}_{=1} \\
&= p^{prev}(z)
\end{aligned}
$$

Extra: Theory of MCMC (3)

- We want our chains to converge (in the limit) to the same stationary state, regardless of starting distribution.
  - Such chains are called **ergodic**.
- For finite state spaces, our chains converge to equilibrium under two relatively weak conditions
  1. **Irreducibility**: Any set of states can be reached from any other state in a finite number of moves $\rightarrow$ Stationary distribution is unique
  2. **Aperiodicity**: The chain does not get trapped in cycles
- Rarely justified theoretically
  - Instead, we can check it empirically by running the estimation multiple times (chains) and checking the convergence (e.g., trace plot / Rhat)

Extra: Metropolis-Hasting (M-H) algorithm

1. **Initialization**: Choose an arbitrary point x'
2. For each iteration at step m,
   2.1 **Propose** a candidate z^* for the next sample by picking from a proposal distribution $q(z^* \mid z^m)$
   2.2 **Accept** the sample if $u \leq \alpha(z^*, z^m)$, where $u \sim \mathrm{Unif}(0, 1)$ and

$$\alpha(z^*, z^m) = \min\left(1, \frac{\tilde{p}(z^*)q(z^m \mid z^*)}{\tilde{p}(z^m)q(z^* \mid z^m)}\right)$$

- This works because it satisfies the detailed balancing:

$$p(\boldsymbol{z})q(\boldsymbol{z} \mid \boldsymbol{z}')\alpha(\boldsymbol{z}', \boldsymbol{z}) = \min\left\{p(\boldsymbol{z})q(\boldsymbol{z} \mid \boldsymbol{z}'), p(\boldsymbol{z}')q(\boldsymbol{z}' \mid \boldsymbol{z})\right\}$$

$$= \min\left\{p(\boldsymbol{z}')q(\boldsymbol{z}' \mid \boldsymbol{z}), p(\boldsymbol{z})q(\boldsymbol{z} \mid \boldsymbol{z}')\right\} = p(\boldsymbol{z}')q(\boldsymbol{z}' \mid \boldsymbol{z})\alpha(\boldsymbol{z}, \boldsymbol{z}')$$

Extra: Metropolis algorithm

- Suppose that your proposal distribution is symmetric: i.e.,

$$q(y \mid x) = q(x \mid y)$$

    - Example: $q(x \mid y) = q((x-y)^2)$

- Then, your ratio is

$$\alpha(z^*, z^m) = \min\left(1, \frac{\tilde{p}(z^*)}{\tilde{p}(z^m)}\right)$$

- This is called **Metropolis algorithm**
- **Problem (both M and M-H)**: The acceptance rate may be low, sensitive to the choice of $q(\cdot \mid \cdot)$, does not work well for high-dimensional settings.

Extra: Gibbs Sampling

- **Setting**: Suppose we want to get $p(z_1, \cdots, z_k)$
- **Algorithm**: After initialization, for each step m, keep sampling

$$
\begin{gathered}
z_1^{m+1} \sim p(z_1 \mid z_2^m, \cdots, z_k^m) \\
z_2^{m+1} \sim p(z_2 \mid z_1^{m+1}, z_3^m, \cdots, z_k^m) \\
\vdots \\
z_k^{m+1} \sim p(z_k \mid z_1^{m+1}, z_2^{m+1}, \cdots, z_{k-1}^{m+1})
\end{gathered}
$$

- **AD**: Always accept the sample
- **DA**: This approach works only when full conditional distributions are easy to sample

Extra: Gibbs Sampling as Special Case of M-H

- If we use the proposal $q(z^* \mid z^m) = p(z_i^* \mid z_{-i}^m)$ and $q(z^m \mid z^*) = p(z_i^m \mid z_{-i}^*)$, then Gibbs Sampling can be seen as a special case of M-H
- In Gibbs Sampling, we always accept the proposal because

$$\begin{aligned} \alpha(z^*, z^m) &= \min\left(1, \frac{\tilde{p}(z^*)q(z^m \mid z^*)}{\tilde{p}(z^m)q(z^* \mid z^m)}\right) \quad \text{(Definition)} \\ &= \min\left(1, \frac{\tilde{p}(z_{-i}^*)\tilde{p}(z_i^* \mid z_{-i}^*)p(z_i^m \mid z_{-i}^*)}{p(z_{-i}^m)\tilde{p}(z_i^m \mid z_{-i}^m)p(z_i^* \mid z_{-i}^m)}\right) \\ &= \min(1, 1) = 1 \end{aligned}$$

where the last equality is by $z_{-i}^* = z_{-i}^m$ (as only i changes)

Extra: Metropolis within Gibbs

- Gibbs sampling is sometimes hard to implement when full-conditional is not known
    - M-H can solve this problem, but does not work well for high-dimensional setting
    - This is because M-H considers the update of all the parameters at the same time as marginal

- We instead consider doing the M-H for each full conditional in Gibbs sampling

- This is called **Metropolis within Gibbs**

Extra: Collapsed Gibbs Sampling

- We can speed up the Gibbs sampler by marginalizing some of the latent variables.
  - This is called **Collapsed Gibbs Sampler**
  - **AD**: Faster convergence (b/c fewer parameters to be sampled)
- **Setup**: let's split the latent variable into
  1. z: variables to sample
  2. θ: variables to collapse
- Then, instead of sample from $p(z, \theta)$, we can integrate out θ:

$$p(z \mid x) = \int p(z, \theta \mid x) d\theta$$

  Then, we run Gibbs sampler updates like:

$$z_i^{(t+1)} \sim p(z_i \mid z_{-i}, x)$$

  under the marginal posterior

Extra: Hamiltonian/Hybrid Monte Carlo (HMC)

- **Idea**: To use momentum to update the states to mitigate the random walk behavior of GS and MH:
  - x: position variable / u: momentum variable
  - Consider

$$\underbrace{H(x_i, u_i)}_{\text{Hamiltonian}} = \underbrace{U(x_i)}_{\text{Potential Energy}} + \underbrace{K(u_i)}_{\text{Kinetic Energy}} = -\log p(x) + \frac{1}{2}u^T M^{-1} u$$

which satisfies Hamiltonian Equations:

$$\frac{\partial x}{\partial t} = \frac{\partial H}{\partial u}, \quad \frac{\partial u}{\partial t} = -\frac{\partial H}{\partial x}$$

Extra: Hamiltonian/Hybrid Monte Carlo (HMC)

- For each iteration at step m,
  1. Updating momentum distribution $u_0 \sim N(0, M)$
  2. For $l = 1, \cdots, L-1$, updating (x, u) by
$$\begin{aligned} u^{l+1/2} &\leftarrow u^l + \frac{1}{2}\epsilon \frac{d \log(x \mid y)}{dx} \\ x^{l+1} &\leftarrow x^l + \epsilon M^{-1} u^l \\ u^{l+1} &\leftarrow u^{l+1/2} + \frac{1}{2}\epsilon \frac{d \log(x \mid y)}{dx} \end{aligned}$$
  3. Accept (u^L, x^L) with probability
$$\min\left(\frac{p(x^* \mid y)p(u^*)}{p(x^{t-1} \mid y)p(u^{t-1})}, 1\right)$$

- **Takeaway**: HMC is one type of M-H methods!
  - Notice that we accept or reject in the same way as M-H